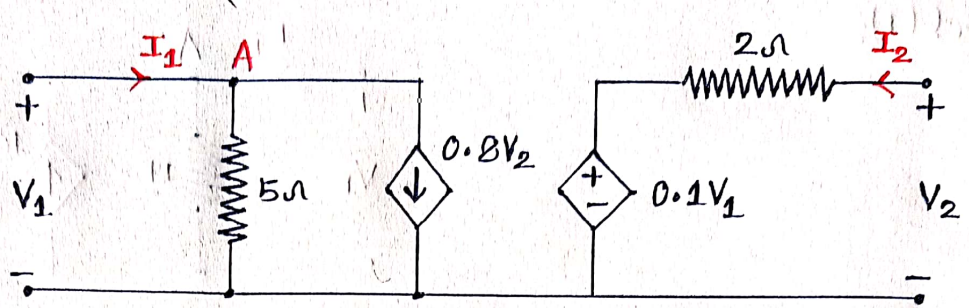


CTD QUIZ - 6 SOLUTION (SET-A)

Sol (1):



Apply KCL at node-A, we get -

$$I_1 = \frac{V_1}{5} + 0.8V_2 \quad \text{--- (1)}$$

Apply KVL at output loop, we get -

$$V_2 = 2I_2 + 0.1V_1 \quad \text{--- (2)}$$

Put the value of V_2 from eqⁿ(2) in eqⁿ(1), we get -

$$I_1 = \frac{V_1}{5} + 0.8(2I_2 + 0.1V_1)$$

$$I_1 = 0.2V_1 + 1.6I_2 + 0.08V_1$$

$$0.28V_1 = I_1 - 1.6I_2$$

$$V_1 = 3.57I_1 - 5.71I_2 \quad \text{--- (3)} \quad \rightarrow (1 \text{ Marks})$$

Put the value of V_1 from eqⁿ(3) in eqⁿ(2), we get -

$$V_2 = 2I_2 + 0.1(3.57I_1 - 5.71I_2)$$

$$V_2 = 2I_2 + 0.357I_1 - 0.571I_2$$

$$V_2 = 0.357I_1 + 1.429I_2 \quad \text{--- (4)} \quad \rightarrow (1 \text{ Marks})$$

As we know that -

$$Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0} = 3.57 \Omega$$

$$Z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0} = 0.357 \Omega$$

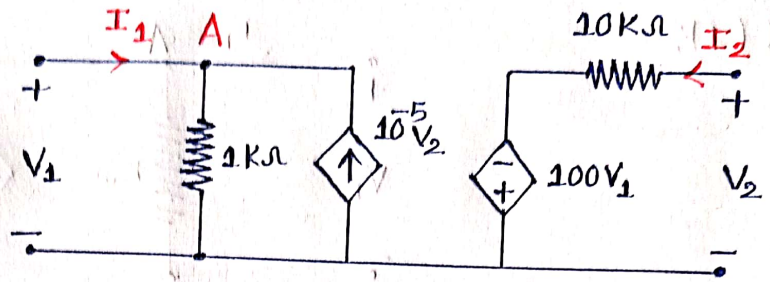
$$Z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0} = -5.71 \Omega$$

$$Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0} = 1.429 \Omega$$

CTD QUIZ-6 SOLUTION

(SET-B)

Sol (1):



Apply KCL at node A, we get —

$$I_1 = \frac{V_1}{1k\Omega} - 10^{-5} V_2$$

$$V_1 = 10^3 I_1 + 10^{-2} V_2$$

— (1)

→ (1 Mark)

Apply KVL in output loop, we get —

$$V_2 = 10k\Omega \times I_2 + 100V_1$$

$$I_2 = \frac{100}{10k} V_1 + \frac{V_2}{10k}$$

$$I_2 = 10^{-2} V_1 + 10^{-4} V_2$$

$$I_2 = (10^3 I_1 + 10^{-2} V_2) 10^{-2} + 10^{-4} V_2$$

$$I_2 = 10 I_1 + 2 \times 10^{-4} V_2$$

— (2)

→ (1 Mark)

As we know that —

$$h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0} = 10^3 \Omega = 1k\Omega$$

→ (2 Marks)

$$h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0} = 10^{-2}$$

→ (2 Marks)

$$h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0} = 10$$

→ (2 Marks)

$$h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0} = 2 \times 10^{-4} \text{ A/V} = 0.2 \text{ mA/V}$$

→ (2 Marks)